

Manual

Detailed Scheduling/ Shop Floor Scheduling HLS-FPL 8.1

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1 Übersicht Feinplanung / Leitstand

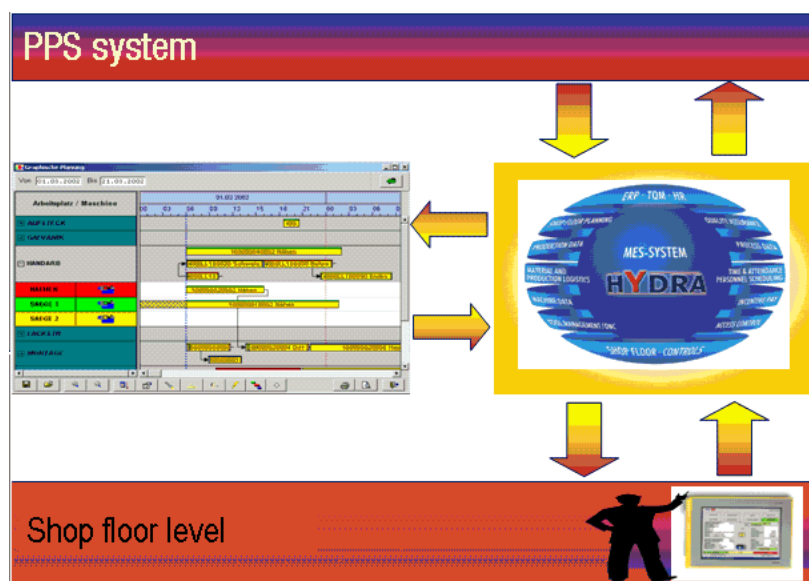
Purpose

Production companies generally use an ERP system to manage their internal resources. This kind of resource management is based on planning data and master data. In this case, current information from the production process is only available in the form of order confirmations in the sense of partial or final confirmations including quantities produced and times accumulated. Specific planning information like availability of personnel, tool availability, machine downtimes or mechanical breakdowns is not available here and can also not be used for daily planning, which is the prerequisite for maximum capacity utilization and therefore, at the same time, a competitive factor.

The information required for daily planning is available in data entry systems (BDE) close to production. Based on this proximity to production, we can conclude that detailed scheduling should be performed as a sub function of production planning and controlling at the "shop floor" level, that is in the data entry system that is close to production.

The data entry system HYDRA shop floor scheduling satisfies this requirement and combines the ERP level with the process level. The basis for detailed scheduling in HYDRA shop floor scheduling is the rough scheduling performed by the ERP system (MRP II run) and the current information from business data entry (BDE).

The following illustration shows HYDRA shop floor scheduling in the system environment of production planning and controlling and also data entry:



In this environment the shop floor scheduling module assumes the following functions:

- Orders are transferred (production, repair and maintenance orders) to the planning board
- Planning is executed based on current process data (availabilities, disturbances, etc.)
- When the operation is scheduled it is given a start and end time (planned start and planned end)
- The schedule is released and automatically transferred to sequencing in the production process
- Planning data is uploaded to the ERP system

Current planning information from business data entry is displayed in HYDRA shop floor scheduling or it impacts the planning situation.

Thus, any disturbances posted or automatically entered in production are indicated as color-coded workplaces/machines in the HLS. In addition, the graphical presentations of order duration are shortened or extended as a result of the input of the latest quantity information. The current order and machine data are retrieved from the HYDRA database and made available in shop floor scheduling, thus depicting the current production situation.

The modifications implemented to order data in HYDRA shop floor scheduling are not transferred to sequencing at the respective workplace until the schedule is confirmed by saving it.

Implementation considerations

You use the function package if:

- You would like to run capacity planning for your workplaces and machines,
- You would like to perform a sequencing within your workplaces and machines.

Integration

The ability to seamlessly integrate HYDRA shop floor scheduling into order, plant and machine data collection allows direct insight into current production results such as order progress, produced yield and scrap quantities as well as machine downtimes – a critical aspect that assures that current planning is always compared to the actual status.

Furthermore, HYDRA shop floor scheduling is linked with the following components, among others, tool and resource management (WRM), for example.

Features

This component provides special basic functions that are accessed by the functions of other components. It

- Calculates remaining runtime

- Visualizes order progress (bar length) by dynamically calculating processing time/remaining run time using predefined formulas. Accounts for the performance level defined at the workplace.
- Accounts for current confirmations
 - Processes current BDE uploads (operation status, quantities entered).
 - Processes current uploads from the MDE (machine states).
- Takes predecessor/successor relationships into account
 - Considers predecessor/successor relationships of linked operations within a multi-level production order (order network).
- Accounts for shift calendars
 - Accounts for the availability of machines and workplaces based on a shift calendar assigned to them.
- Performs automatic assignment
 - Automatically dispatches the operations transferred from the ERP system taking into account planning strategies and/or priority rules.
- Calculates key performance indicators
 - Calculates key performance indicators used to assess scheduling (e.g. total number of delays, setup times, machine utilization).
- Is an order / operation pool
 - Stages an order / operation pool with reference to deadlines and durations
- Is a planning profile
 - Takes various planning profiles into account.

2 Planning Profiles

Overview

Menu	Master data → Production control → Planning profiles
Transaction code	Plprof
Function authorization	Plprof

Purpose

You use this function to create or modify planning profiles in the system.

Integration

By using planning profiles, you can narrow down the data displayed in the different planning functions (e.g. workplaces, staff).

- [Graphic planning](#) (transaction grap)
- Graphic order sequencing (transaction graps)
- [Workplace assignment](#) (transaction wpas)

Requirements

You have structured the workplaces to be planned based on capacity groups and set them up in the system.

Selection criteria

The application provides the following selection criteria:

User

User name for whom the planning profiles that were configured beforehand are to be displayed. You can also run a search using wildcards.

Global planning profiles are not shown if you select a specific user.

Planning profile

Name of the planning profile to be searched. You can also run a search using wildcards.

Field descriptions

User

User for whom the subsequent capacity group is to be assigned or has been assigned in the planning profile.



You can leave this field empty if the modification [PLPROF-GLOBAL](#) has been enabled.
This is a global planning profile that can be selected for every user.

Profile

Name of the planning profile.

You cannot use names in global and user-specific planning profiles.

Selection

You can assign different objects to a planning profile subject to the application and its product version:

Application	Modification	
Graphic planning		Group
Graphic planning	PLPROF-MNR	Group, workplace  If you assign single workplaces to a planning profile, the dialog still shows the group the workplaces belong to.
Graphic order sequencing		Group
Workplace assignment		For workplaces: group For staff: area, cost center, department, employee subgroup, employment relationship, person
Workplace assignment	PLPROF-MNR	For workplaces: workplace, group For staff: area, cost center, department, employee subgroup, employment relationship, person

Value

Subject to the selection made, you have to enter the relevant value in this field.

If you select "group", it might be a capacity group configured as bottleneck or throughput capacity.

In general, the graphic planning board, graphic order sequencing or workplace assignment only shows the workplaces for which you are authorized via the responsibility.

Order

This field specifies the order of groups in the [graphic planning board](#).



We strongly advise to input data in this field, because the order of groups might otherwise be arbitrary. Ideally, you should use intervals of 10.



If the modification [PLPROF-MNR](#) is enabled, the value defined for the workplaces specifies the order of groups when you assign workplaces to a planning profile.

Within a group, you can sort workplaces by using the “position” field of the [group assignment](#).

The following options are only relevant for the [graphic planning board](#):

Visible in shop floor planning

This field specifies if the current group is shown when selecting a planning profile.

Show workplaces without responsibility area authorization

This option shows the workplaces of the selected group in the Shop Floor Scheduling module although the user does not have the required responsibility area authorization.

Allow planning without responsibility area authorization

When resolving conflicts occurred during manual planning, this option moves an operation for which the user does not have the responsibility area authorization. You cannot shift the operation manually; rather shifting can only be done as a means of resolving conflicts.

3 Planning Variants

Overview

Menu	Master data → Production control → Planning variants
Transaction code	plvar
Function authorization	mdplvar

Purpose

Use this function to create or modify planning variants in the system.

Integration

A planning variant combines specific settings that are relevant for the planning. The settings included in a planning variant are usually used by the function *Automatic assignment*.

Requirements

The [Priority rule](#) used is defined in the system.

Selection criteria

The application provides the following selection criteria:

Planning variant

Enter the required search criterion to search for a planning variant. Please note case sensitivity.
You can also use wildcards.

Personal

This checkbox is a "tri-state" checkbox that can have three different states:

- ☐ Only those planning variants are selected that are not identified as *personal*.
- ☒ Only those planning variants are selected that are identified as *personal*.
- ☐ All planning variants are selected.

Field descriptions

Planning variant

Name of the planning variant

Comment

Comment

Owner

User name of the owner of the variant.

Personal

If a planning variant is identified as personal, only the person who created this variant can use or edit it.

Planning lead time

Period of time that the planning function identifies as lead time; result: no assignment is made. The period of time specified is the minimum time that is required to implement a planning.

Planning time fence

After an automatic scheduling run, the system automatically fixes the operations that start in the specified planning time fence if the relevant option is enabled in the [HLS settings](#). This means that the planning time fence must not be larger than the planning horizon.

Priority rule

You can select a strategy for the automatic planning. You can select from preconfigured rules (HLS-RBM) and self-configured rules (HLS-VMB; HLS-ZMB). The following preconfigured rules are available:

- Lowest setup costs: to calculate the setup costs, setup time + teardown time are totaled (dynamic setup times are not integrated).
- Shortest operation time: operation time = setup time + processing time + teardown time
- Shortest remaining processing time: the remaining processing time is calculated using the remaining run time formula stored for the OP.
- Longest operation time: operation time = setup time + processing time + teardown time
- Longest remaining processing time: the remaining processing time is calculated using the remaining run time formula stored for the OP.

Displacement to the left

In case of an automatic planning of operations, this option can specify different start dates for the scheduling. Condition: The start of the operation must be within the planning horizon.

If this option is disabled, then the scheduled start date will be the earliest possible date (forward scheduling) or the latest possible start date (backward scheduling).

If the option is enabled, the "left shift" is activated and the scheduled start date is set to "now" plus planning lead time. In other words, an attempt is made to plan the operation as early as possible.



This option ignores the scheduling result, i.e. you can plan the operation before the earliest start date. If the scheduling is performed in the higher-level system, we recommend to disable this option.

Capacity selection

The *Capacity selection* controls for which capacity, i.e. workplace, the operation is planned that is next in the queue and must be planned next.

The first free capacity

The capacity that is available first in terms of time.

Capacity having the most "capacity"

Selects the capacity that provides the most available capacity with respect to the planning horizon.

Capacity having the least "capacity"

Selects the capacity that provides the least available capacity with respect to the planning horizon.

Shortest processing time

Selects the capacity where the processing time is the shortest (calculation includes the performance level).

Note: This is important with production variants because the values to calculate the process times can change. The capacity that is available first in terms of time.

Shortest total setup time (static + dynamic)

This rule for capacity selection integrates the static and dynamic setup time when assigning operations to a workplace. If this option is set, the system always assigns the operation to the workplace where the total of static and dynamic setup time is the lowest. If two machines have identical setup times, the system selects the workplace with the number that comes first in alphanumeric order.

Shortest static setup time

Selects the capacity where the static setup time is the shortest.

Highest priority

If this option is set, this rule for capacity selection uses the priority of the production variant (assigned to the relevant workplace) to assign an operation to a workplace.

If you do not use production variants, all workplaces have the same priority. The system selects the workplace that provides available capacities first. If two workplaces (production variants) have the same priority, the system selects the workplace that provides available capacities first. If two workplaces provide available capacities at the same time, the system selects the workplace with the number that comes first in alphanumeric order. If there are two production variants using different tools, the system selects the production variant using the tool with the number that comes first in alphanumeric order.

Note: The higher the priority value of a production variant, the greater the probability that this variant is used.

4 Individual shift/assignment times

Overview

Menu	Production control → Preparations for production → Individual shift/assignment times
Transaction code	mdistmf
Function authorization	mdistmf.*

Purpose

You can define individual shift/assignment times within the [Graphic planning](#) and the [Workplace assignment](#) and specify for a workplace within a specific period of time, whether this time is to be considered as working time or idle time.

This enables short-term modifications with respect to the availability of workplaces, without having to change the planned shift model.

This application manages times without shift or shift times that have been assigned to one or several workplaces within the functions Graphic planning or Workplace assignment.

Integration

Normally, individual shift/assignment times are directly defined by the corresponding functionality provided in the Graphic planning or the Workplace assignment.



These individual shift/assignment times do not affect collection and posting within the scope of shop floor data collection.

Requirements

The product group Graphic planning or Workplace assignment is in use.

Selection criteria

The application provides the following selection criteria:

Group

You can restrict the entries to a specific group by using this combo box.

Workplace

Enter a workplace to view only entries for this workplace.

Period from

Optionally, you can use this input field to enter the beginning of a period as of which you want to display the entries in the application.

Working time

This option selects

- ☐ times without shift only
- ☒ working times only
- ☐ both times



Please note that 3 states are available with this checkbox. If you require data that is missing, have a look at the checkbox setting. We recommend to set the checkbox to the option ☐.

Active

You can enable/disable individual shift/assignment times. The Graphic planning does not include disabled entries.



Please note that 3 states are available with this checkbox. If you require data that is missing, have a look at the checkbox setting. We recommend to set the checkbox to the option ☐.

Field descriptions

Workplace

Workplace for which the entry (individual shift/assignment time) has been created.

Group

Group of the workplace for which the entry (individual shift/assignment time) has been created.

Period from

Point in time when the individual shift/assignment time begins.

Period until

Point in time when the individual shift/assignment time ends.

Working time

This option describes, if it is

- ☐ a time without shift
- ☒ working time.

If it is not working time, but a time without shift, the workplace capacity is not available during this period. Therefore, you cannot plan an operation.

Active

You can enable/disable individual shift/assignment times. The Graphic planning does not include disabled entries.

We recommend to set this option in general to .

Comment

You can store a short comment for this individual shift time including further details.



In the Graphic planning, you can additionally specify a color for the time without shift when defining an individual shift/assignment time.

In the Graphic planning, you can display the comment that is stored for an individual shift in the tooltip of the corresponding individual shift (as of HLS 8.2). Enable (disable) the tooltip using an INI data entry.

- Name: HLS
- Section: SCHEDULING
- Key: DISPLAY_TOOLTIP_FOR_ISTMF
- Value: J
- Active: [selected]

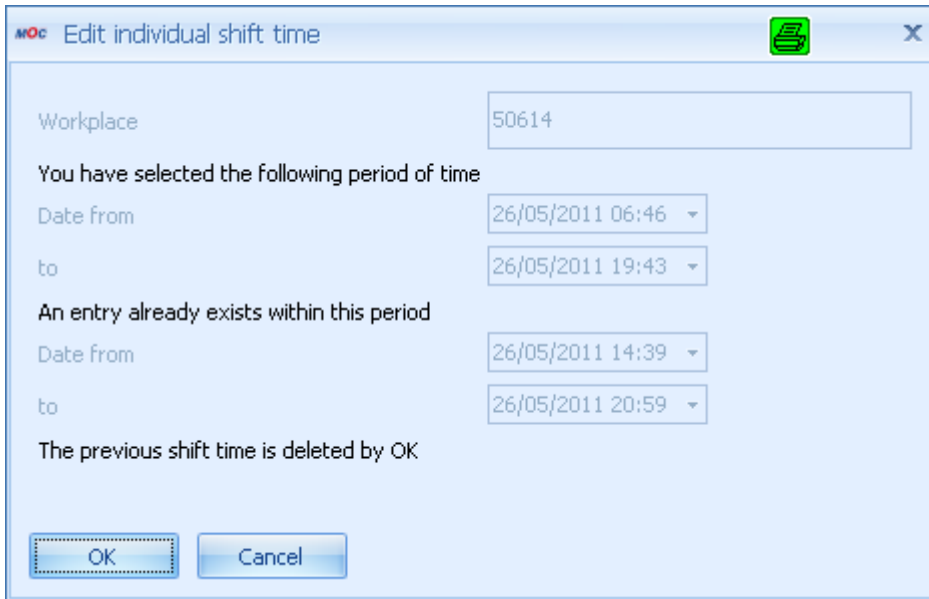


If you enable the tooltip display, the presentation color of times without shift changes in the Graphic planning. In addition, the workplace bars are displayed one level before the times without shift.

Editing functions

Use the available buttons to create or edit individual shift/assignment times.

Only one entry can exist at a workplace for each period. If you make an entry and the system detects that an entry already exists for this period, you can either delete the previous entry or cancel this entry.



If multiple individual shift times exist, this dialog is opened for each existing shift time, and the user can decide whether or not to delete the entries.



Please be careful not to define overlapping periods for one workplace.

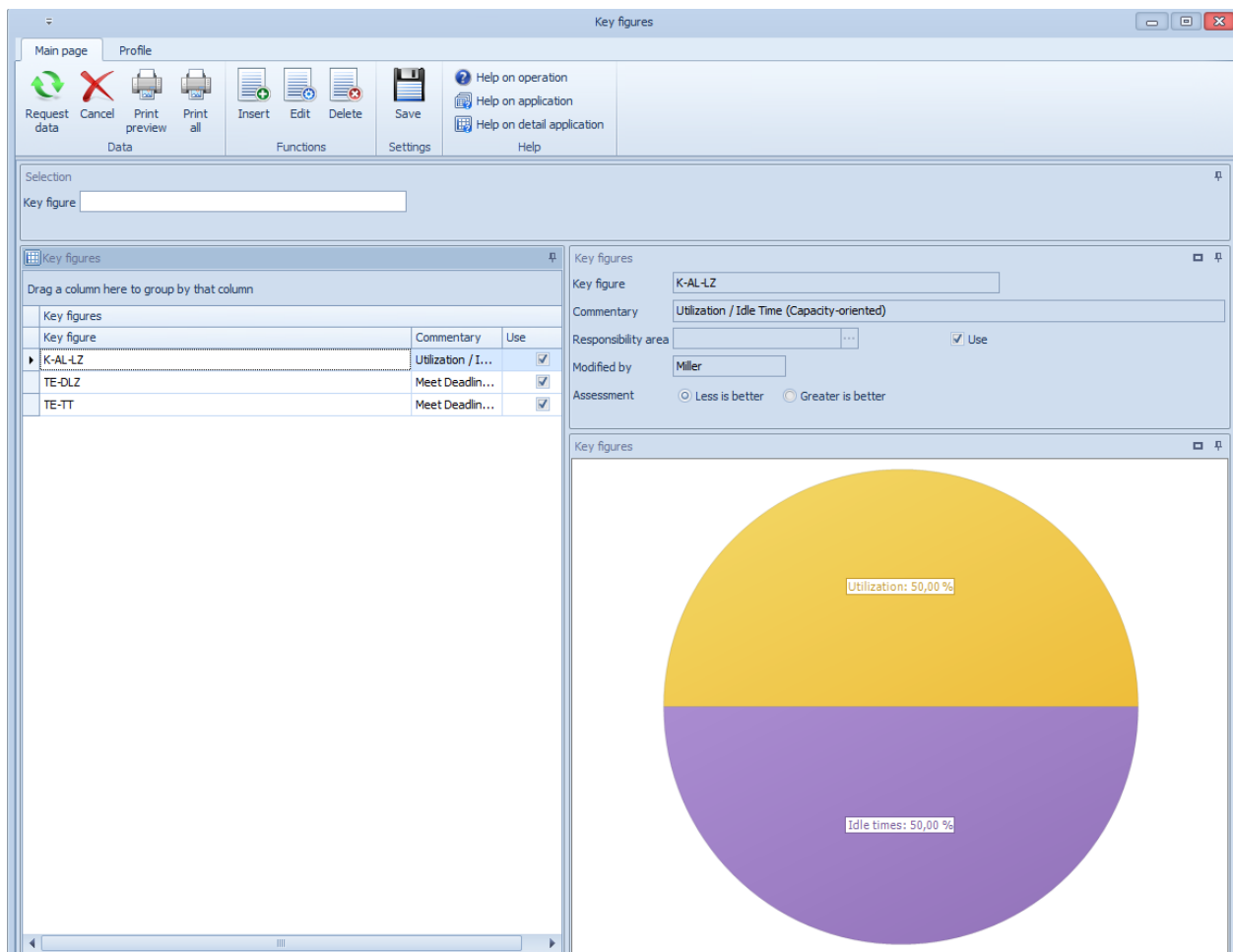
These additional shift times are only used in Shop Floor Scheduling, not as part of data collection.

The user can only change, view and delete machines that belong to the responsibility area he/she is authorized for.

5 KPI

Overview

Menu	Master data → Production control → Key figures
Transaction code	plkeyf
Function authorization	mdplkeyf



The screenshot displays the SAP 'Key figures' configuration window. The left pane shows a list of key figures with columns for 'Key figure', 'Commentary', and 'Use'. The right pane shows the configuration for the selected key figure 'K-AL-LZ', including a commentary 'Utilization / Idle Time (Capacity-oriented)', a responsibility area, and a modified by field. Below the configuration fields is a pie chart showing the distribution of utilization and idle times.

Key figure	Commentary	Use
K-AL-LZ	Utilization / I...	☒
TE-DLZ	Meet Deadlin...	☒
TE-TT	Meet Deadlin...	☒

Key figures configuration details:

- Key figure: K-AL-LZ
- Commentary: Utilization / Idle Time (Capacity-oriented)
- Responsibility area: ... ☒ Use
- Modified by: Miller
- Assessment: ☐ Less is better ☐ Greater is better

Pie chart data:

- Utilization: 50,00 % (Yellow)
- Idle times: 50,00 % (Purple)

Purpose

You use the function *Key figures* if you want to define further key figures that are based on the basic key figures available in the system.

Integration

You can use key figures to evaluate an existing planning.

The individual key figures are available in the *Graphic planning* if the option *Use* is enabled for the respective key figure. For a description on how the key figures are shown in the Shop Floor Scheduling module, [here](#).

When the different planning versions are saved, the key figures are also saved. You can later on compare the saved planning versions with respect to the key figures.

Selection criteria

The application provides the following selection criterion:

KPI

Configured key figure. You can also use wildcards.

Field descriptions

KPI

Name of the key figure

Comment

Text field including further information on the key figure.

Use

If the option *Use* is set, the key figure is available in the *Graphic planning*. When the planning is saved, the key figure is saved with the planning (*Save planning as*).

Responsibility area

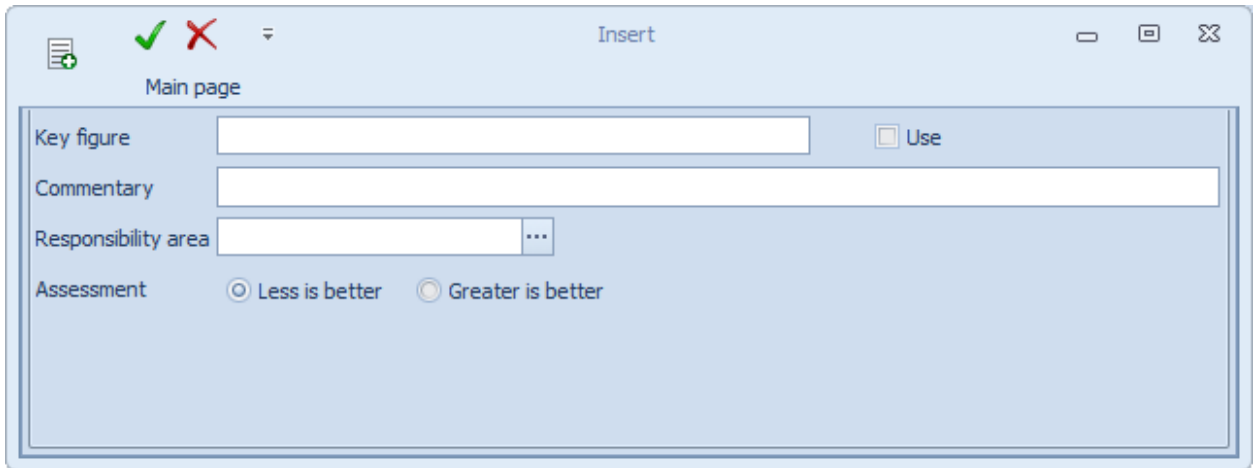
You use the responsibility area to control the access to the key figure.

Note: This field is not available in the table.

Assessment

This option specifies if a calculated key figure is better if its value is smaller (e.g. setup costs) or if its value is greater (e.g. utilization). This setting is used in the application *Saved planning* to specify the respective color.

Note: This field is not available in the table.



Editing function

To create a new key figure, proceed as follows:

1. Click the button *Insert* to call the dialog.
2. Enter the required data in the dialog.
3. Save the dialog.
4. Select the new entry in the table and click the button *Edit* to call the editing dialog.
5. Specify a weighting for each basic key figure, which is used to calculate the new key figure. Save the dialog. The basic key figures and their calculation are described in the following paragraph.

Note: When you have created or edited a key figure, you must reload the planning board.

The weightings are without dimension, but internally they are scaled to 100%. Examples:

1. Utilization, value = 50
Setup costs, value = 50. Both key figures are each weighted with 50%.
2. Utilization, value = 1
Setup costs, value = 3

The utilization is weighted with 25% and the setup costs with 75%.

Key figure

TE-DLZ

Use

Responsibility area

Commentary

Meet Deadlines / Throughput Time

Assessment

☒ Less is better
 ☐ Greater is better

Modified by

Miller

Modified on

21.09.2014 17:18:35

Key figure	Value of weighting factor
Capacity-oriented	
Idle times	0
Evaluated idle times	0
Setup costs	0
Labor utilization	0
Evaluated labor utilization	0
Delays (capacity)	0
Utilization	0
Evaluated utilization	0
Order-oriented	
Lead time	30
Wait times	20
Delays	50
Earliness	0

OK

Cancel

Basic key figures

You can use key figures to evaluate a planning. To evaluate a planning, the system only uses the workplaces that are loaded in the current planning scenario.

By default, HYDRA provides the following key figures (basic key figures). If you create your own key figures, you can use the majority of these basic key figures.



Planned start and planned end of an operation are always recalculated in the planning board. The times can be different to the entries "planned start/end" in the order information dialog.

Capacity key figures

Overall capacity

Total of the durations of all operations that can be planned for a machine/workplace in the specified planning horizon (maximum number). The parameters, which specify the number of possible operations, are defined in the workplace configuration (workplace type, year model, logon of several OPs, etc.).

Usage

Total of the occupancy times of all operations in the planning horizon. The occupancy time of an operation is identified as follows: planned end - planned start

Utilization rate

Ratio of the key figures "Utilization" and "Overall capacity". The utilization rate specifies the percentage of time a machine/workplace is utilized.

Evaluated utilization

Total of the occupancy times of all workplaces multiplied by the standard rate of the machine. The standard rate of the machine is defined in the machine/workplace configuration.

Idle times

Total of all idle times of all workplaces. The idle times are the working times defined in the shift model when no operation is planned.

Evaluated idle times

Total of the idle times of all workplaces multiplied by the standard rate of the machine. The standard rate of the machine is defined in the machine/workplace configuration.

Delays (capacity)

Total of all delay times of the operations in the planning horizon.

Calculation of the delay time of an operation:

If $LET - \text{planned end (resulting from planning)} < 0$

then the delay time = planned end - LET, with

LET = Latest end date/time of the operation.

Setup costs

Total of the static setup times, teardown times and the dynamic setup times of all operations.

To integrate dynamic setup times, you require the respective license.

Setup rate

Ratio of the key figures "Setup costs" and "Utilization". The setup ratio specifies the proportion of the setup costs in the utilization of a machine.

Labor utilization

Product of the workforce requirement, which is stored in the machine operator relation "production", multiplied by the processing time of the operation, summed up for all operations loaded. The machine operator relation (M/O rel.) is stored in the operation (see Order information → tab Processing → M/O rel. product.). The processing time is specified in the respective entry stored for the operation.

Evaluated labor utilization

Product of the workforce requirement, which is stored in the machine operator relation "production", multiplied by the processing time of the operation and the standard labor rate, summed up for all operations loaded. The standard labor rate of an operation is defined in the machine/workplace configuration.

Order key figures**Delays (order)**

Total of all delayed orders (time) loaded in the planning horizon.

Calculation of the delay time of an order:

If basic end date of the order header - planned end of the last OP < 0

then delay time = planned end of the last OP - basic end date of the order header

Earliness (order)

Total of all orders finished too early (time) loaded in the planning horizon.

Calculation of the early finish time of an order:

planned end of the last OP - basic end date of the order header

Wait times (order)

Total of idle times between OPs of one order in the planning horizon.

Calculation of the wait times:

The system uses the total of durations *between* the scheduled OPs of an order: planned start of subsequent OP - (planned end of OP + wait time of OP)

Lead times (order)

Lead times of all resources of an order within the planning horizon.

Planned end of last OP of an order - planned start of first OP of the order.

Total processing time

The total processing time is the time between the planned start of the first operation and the planned finish of the last operation dispatched in the planning scenario.

For example, the key figure " Total processing time" can be used to compare the planning results of automatic occupancy and cognitive planning. In each planning profile, the total processing duration achieved with the automatic assignment function is greater than after cognitive planning.

KPI Total processing duration

6 Automatic Assignment

Purpose

You can use the function *Automatic assignment* to automatically perform a planning scenario, which is defined via planning profile and planning horizon.

The function is used to plan orders and the included operations to capacities. The system uses assessment criteria – the so-called *Assignment strategy* – to sort the operations and to distribute the operations to the workplaces of the group.

Requirements

Before you start the detailed planning, you must define and create the relevant [Planning profiles](#) that integrate aspects of organization, planning procedure and competencies.

6.1 Automatic assignment

The automatic assignment, also called capacity planning, is used to plan orders or their operations to capacities. To meet the demand of each operation, the system selects a resource or a combination of resources from the resources available. As a result of planning, the operation is assigned a start and end time (planned start and planned end).

To calculate the planned duration of the operation, the system uses the workplace/machine that meets the main demand plus the dynamic setup time between the preceding operation and the operation itself.

The workplace/machine is the resource that meets the "main demand". The alternatives to meet the demand are the workplaces/machines of the same group (the group the operation has been assigned to). The assignment of a workplace to a group is configured in the master data.

Production resources and tools are used to meet the further demand. They are stored for the operation.

If [Production variants](#) are used, the combination of workplace and resource meets the relevant demand. Here, a resource is specified in the Tool and Resource Management (WRM) and assigned to a production variant.

An automatic planning run is performed in the following steps:

1. The order of assignment is specified via sorting of the operations that must be planned ("queue") according to the defined **planning strategy** (*priority rule*).

The orders that must be planned are sorted according to their priority and placed in a queue. To identify the priorities, different rules are available.

2. The operations are distributed among the available workplaces according to the specified procedure (**control of capacity selection**).

The following rule applies to **specify the planned date** of an operation that has been assigned to a machine via capacity selection:

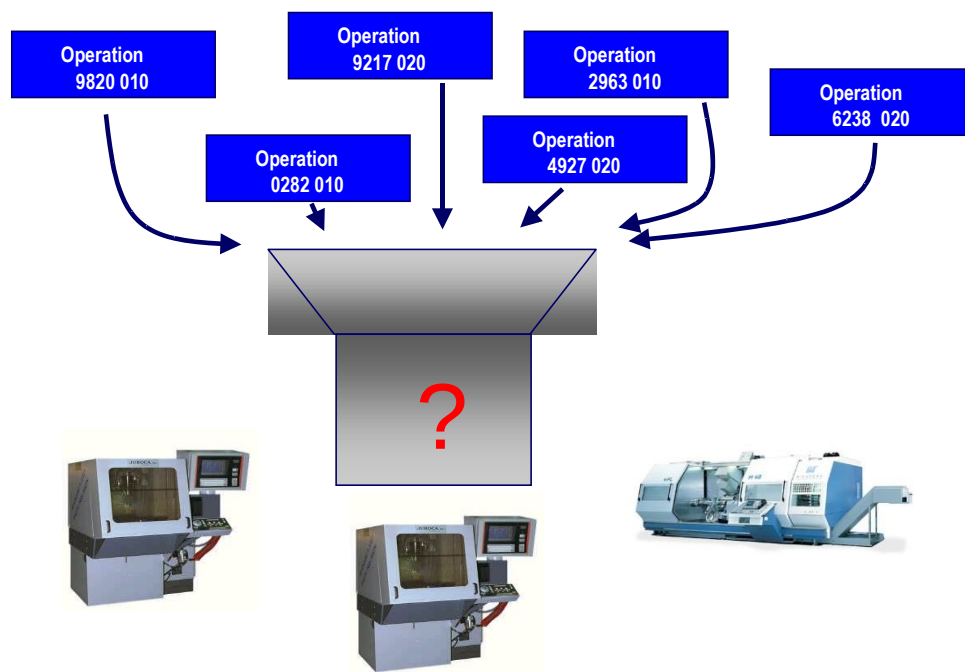
- If the planning variant includes the active option *Displacem. to the left* (left shift) for the planning horizon, then the earliest possible date within the planning horizon is selected as planned start. With this planned start, the capacity is available and the relationships between operations are respected and free of conflicts. - If the option *Displacem. to the left* (left shift) is not active in the planning variant, the system tries to schedule the operation at its **scheduled start time**. In case of a forward scheduling, the start time is the **earliest start**, in case of a backward scheduling, it is the **latest start** of the operation. If no capacity is available at that time, the next later point in time is identified when the required capacity is available. With backward scheduling, the operation is then scheduled "too late". Optionally, the option IGNORE_TIMING_DIRECTION can be set to "J" in the INI configuration. With this configuration and in case of backward scheduling, the automatic planning searches for a start date beginning with the **earliest start**. To identify the start date, the planned machine must provide the required capacity.

The result of the automatic assignment is a planning scenario that can actually be realized. This means that the planning is free of capacity overloads or conflicts because of relationships. If these conditions must be fulfilled, it might not be possible to schedule all operations in due time. It is therefore possible that some operations must be planned with a delay. This depends on the planning situation, the available resources and the selected planning strategy. If the capacity required for an operation is not available until the end of the planning horizon and with respect of the situation and the respective configuration, then this operation remains in the pool of groups also after an automatic assignment.

6.1.1 Specification of the assignment order

The automatic assignment is based on a queue concept. The workplaces assigned to a capacity group are regarded as a unit, in front of which a queue is formed, i.e. the operations to be planned line up in front of this unit. Each queue is given a priority rule to identify the highest priority operation (so-called contender) in the queue.

The user can specifically set queue sorting (assignment strategy) when the system is customized and therefore adapt it to the respective requirements. Each operation is weighted with the corresponding factors based on the assignment strategy. The operations are then sorted accordingly and then distributed to the workplaces.

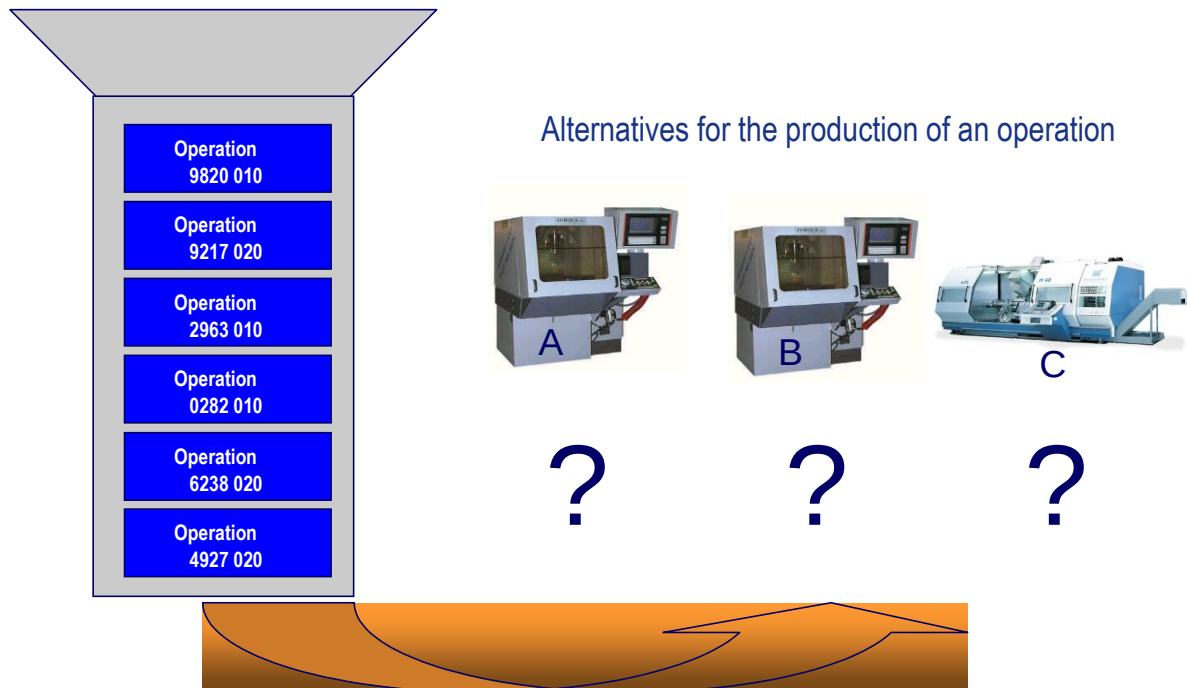


Various bottleneck and scheduling resources can be defined for scheduling within the planning horizon (the specification is made in the [group](#)). When you schedule operations within the planning horizon, resources are checked, i.e. the system checks availability of all resources not only of workplaces.

Apart from the required demand and the possible elements that meet the demand, capacity planning also takes into account further restrictions, such as maximum time intervals between two operations to be synchronized.

6.1.2 Controlling capacity selection

How operations are distributed to the individual workplaces can be controlled using the adjustable capacity selection in the [planning variants](#).



The following alternatives are available:

The first free capacity

The capacity that is available first in terms of time.

Capacity having the most "capacity"

Selects the capacity that provides the most available capacity with respect to the planning horizon.

Capacity having the least "capacity"

The capacity is selected which, with respect to the planning horizon, provides the least capacity.

Shortest processing time

When attempting to dispatch operations, the order in which capacities are selected is based on the shortest processing time (while taking into account performance level).

Please note: This is of particular significance in conjunction with alternative production methods (production variants), since these provide certain values used in calculating process times.

Shortest setup time

When attempting to dispatch operations, the order in which capacities are selected is based on the shortest setup time. Dynamic setup times based on the increases or reductions in setup time defined in the [setup change matrix](#) are taken into consideration.

Constraints within the order network are appropriately taken into consideration during automatic scheduling. Operations that are already running are not rescheduled. If an assignment is not possible because there are overlaps with the predecessor or successor operations, for example, the operation is not scheduled and appears in a list at the end of automatic planning.

During automatic assignment, fixed operations that are still set in the past are moved to the right and set to "now" plus planning lead time at the earliest. Fixed operations in the future are not modified, but remain scheduled as they are.

6.1.3 Complete assignment of planning board



Automatic assignment of the entire planning board is accessed using the icon in the toolbar. A dialog opens from which the following options can be selected:

Schedule unplanned operations only (pre-assignment)

Using this option operations that are in the pool of groups are weighted according to the current assignment strategy and distributed to/ scheduled for the workplaces of the group. Operations that have already been scheduled remain scheduled in their position (workplace, date).

If an assignment is not possible because there are overlaps with the predecessor or successor operations, for example, the operation is not scheduled and appears in a list at the end of automatic planning.

Reschedule entire planning scenario

If all operations are to be rescheduled, select option "Reschedule all". In this case, all workplaces for operations already scheduled, and that are not fixed or running, are scheduled first, that is moved to the pool of groups before automatic assignment takes place.

After this has been confirmed, the separate workplaces are now assigned. In the process, all operations in the pool of groups are weighted according to the current assignment strategy. These operations are then distributed/ dispatched to the workplaces in the group.

If an assignment is not possible because there are overlaps with the predecessor or successor operations, for example, the operation is not dispatched and appears in a list at the end of automatic planning.

6.2 Planning strategies/priority rules

A planning strategy/ priority rule specifies a priority code (PKZ) for each of the operations transferred to the queue.

In addition to the predefined strategies in the system - rule-based machine assignment - user-defined strategies are also possible (target-oriented rules and sorting). Whether or not individual planning algorithms can be used for this purpose depends on the license. Planning algorithms are selected through selecting a planning variant.

6.2.1 Rule-based machine scheduling

Lowest setup costs

The setup costs equal the sum of setup time + teardown time.

The operation in the queue having the lowest setup costs for the machine in question gets the highest priority (dynamic setup times are not included; these times are used with the capacity selection).

Shortest operation time

Operation time = setup time + processing time + machine teardown time

The operation in the queue having the shortest operation time gets the highest priority.

Shortest remaining processing time

The operation in the queue having the shortest remaining processing time gets the highest priority.

The remaining processing time is calculated using the remaining run time formula stored for the operation.

Longest operation time

Operation time = setup time + processing time + machine teardown time

The operation in the queue having the longest operation time gets the highest priority.

Longest remaining processing time

The operation in the queue having the longest remaining processing time gets the highest priority.

The remaining processing time is calculated using the remaining run time formula stored for the operation.

6.2.2 Variable machine assignment

Planning strategy for controlling automatic machine assignment based on customer-specific sorting criteria (sorting rules).

The sorting rule calculates a priority number for each of the operations presented. It works more or less like a sifter in which ultimately only the highest priority operation in the queue is trapped. If, for example, only one operation is left over after the first sorting criterion is processed, this operation is scheduled as the next one and the other sorting criteria are not further considered.

The configuration used to create sorting rules is described [here](#). The possible sorting criteria are also described there.

By default, automatic assignment is carried out in the following sequence:

	Sequence	Sorting	
1	Scheduled start (date) of operation	Ascending	
2	Priority of operation	Descending	
3	Order buffer time (from scheduling)	Ascending	
4	Order index of the order	Descending	
5	HYDRA order number (order/OP number)	Ascending	

6.2.3 Target-oriented machine assignment

Planning strategy for controlling automatic machine assignment based on weighted targets.

This planning strategy involves indexing the operations to be scheduled according to a variable weighting and then sorting the queue according to this index. In contrast to priority rules, the actual target is weighted in this case, and assignment is then aligned with the target.

The configuration used to create sorting rules is described [here](#). The possible sorting criteria are also described there.

6.3 Other notes

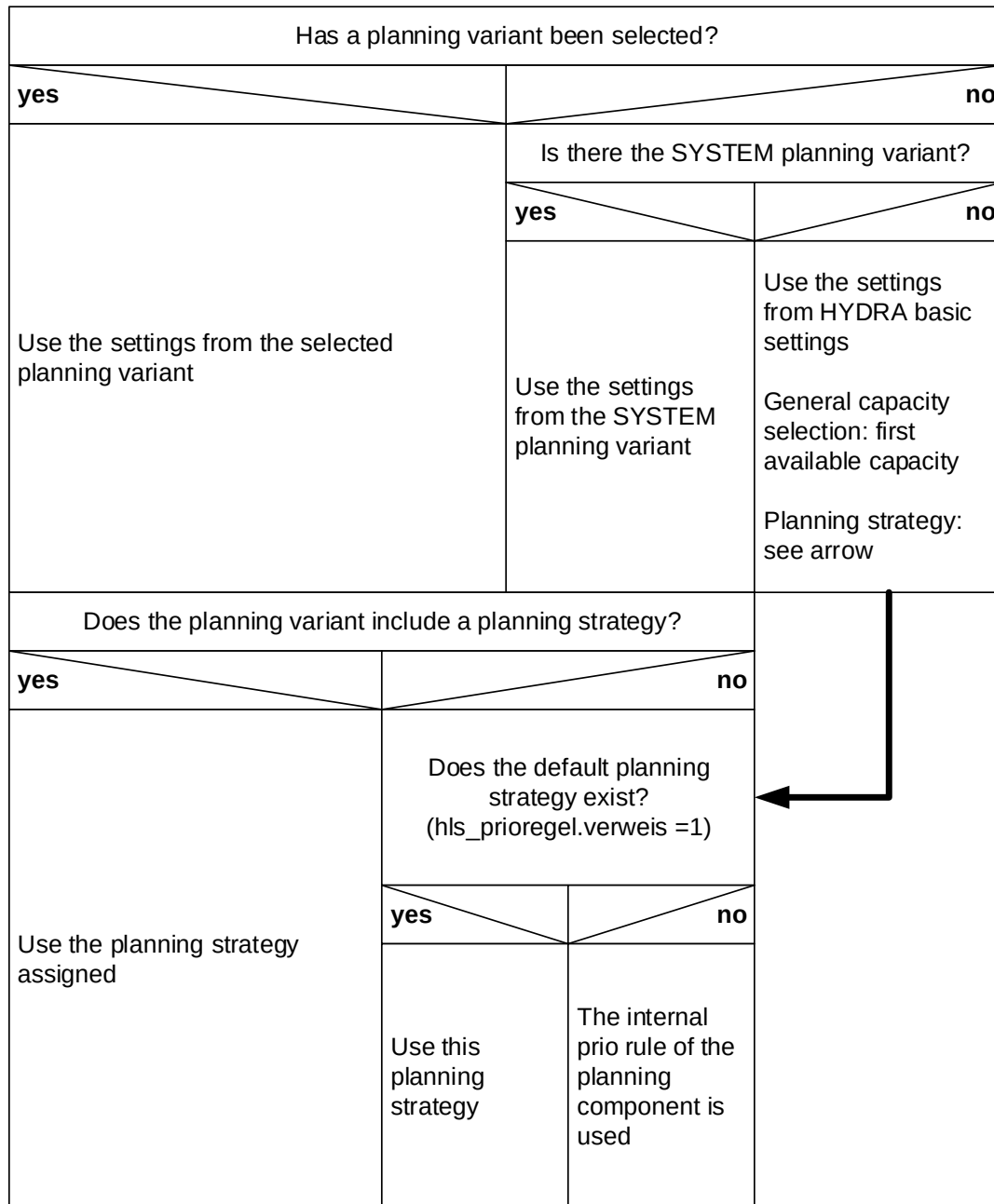
Please note the following:

Taking wait time into consideration

The wait time defined in the operation is NOT taken into consideration in planning (automatic assignment or manual assignment). This wait time is considered being a wait time for planning purposes that is thus only considered as buffer time when it comes to scheduling the lead time.

Planning variants and basic settings

You can make different settings for the time horizons and the automatic assignment in the *Basic settings* and in the *Planning variants*. If you do not select a planning variant in the *Graphic planning*, the basic settings are used. The following graphic shows the logic applied for the use of the different settings.



The internal priority rule (sorting rule) of the planning component is as follows:

	Sequence	Sorting	
1	Scheduled start (date) of operation	Ascending	
2	Priority of operation	Descending	
3	Order buffer time (from scheduling)	Ascending	
4	Order index of the order	Descending	
5	HYDRA order number (order/OP number)	Ascending	

7 Scheduling

Summary

Menu	Production Control → Production Preparation → Scheduling
Transaction code	orterm
Function authorization	or.terminate

Usage

You use the lead time scheduling function if you are set up to use HYDRA shop floor scheduling and your PPS/ ERP system does not allow for lead time scheduling to calculate basic operation-related dates.

Integration

[Lead time scheduling](#) calculates basic operation-related dates based on basic order-related dates – earliest start, earliest end, latest start, latest end. Operations are scheduled between these dates in HYDRA shop floor planning so as to assure that the basic order-related finish date is maintained.

Requirement

The requirements are described [here](#).

Performing scheduling manually

Scheduling can also be initiated manually from the following functions.

- Scheduling can be initiated manually for an order using the  icon in the order information.
- Scheduling can be initiated for all orders in the order backlog from the menu Production control > Production preparation > Scheduling.

The option *Schedule orders modified last* schedules those orders that are pending for scheduling in the table described above, whereas the option *Schedule all orders* will schedule all orders.

8 Lead Time Scheduling

Usage

You use the lead time scheduling function if you are set up to use HYDRA shop floor scheduling and your PPS/ ERP system does not allow for lead time scheduling to calculate operation-related basic dates.

Integration

Lead time scheduling calculates operation-related basic dates based on order-related basic dates – earliest start, earliest end, latest start, latest end. Operations are scheduled between these dates in HYDRA shop floor planning so as to assure that the order-related basic finish date is maintained.

Requirement

The following requirements must be met in order for orders to be scheduled by lead time scheduling:

- The [order type](#) can either be relevant for scheduling or relevant for detailed planning (customized setting).
- The [order status](#) can either be relevant for scheduling or relevant for detailed planning (customized setting).
- The [operation status](#) can either be relevant for scheduling or relevant for detailed planning (customized setting).
- The operation is flagged based on its [processing code](#) as either relevant for scheduling or relevant for detailed planning (customized setting).



This document will provide a description of which order types and status, or which processing code, are flagged as relevant for scheduling or relevant for detailed planning at the time of delivery.

- Scheduling is activated as a cyclic job in the scheduler (Script *hls_term.scr*).

8.1 Procedure

Lead time scheduling determines the start times and/or end times for all operations of an order and consequently also for the order itself. Lead time scheduling includes both forward and backward scheduling. In the operation, forward scheduling determines the earliest start (earliest start time/EST) and the earliest end (earliest end time/EET) while backwards scheduling determines the latest start (latest start time/LST) and the latest end (latest end time/LET).

Defining the EST/EET

In order to define EST/EET, you have the option to specify the production order start date; otherwise the current time will be set. However, the planning lead time defined in the HYDRA basic settings is also included in the calculation of the current time so that the earliest start date is "now" + the planning lead time. When calculating the dates, the system takes the so called factory calendar (i.e. the shift calendar that is designated as "factory calendar") into account.

Example

The following shifts are defined in the factory calendar: 5:45 a.m. - 8:00 a.m., 8:15 a.m. - 11:30 a.m., 12:00 noon - 1:45 pm, ...; in the HYDRA basic settings 5 hours of planning lead time are set.

Start of scheduling: 10:20 a.m.

Calculation: 10:20 a.m. + 5 hours + 0:30 hours for shift break (11:30 a.m.-12:00 noon)

Results: The earliest start time for scheduling is therefore 3:50 pm

Defining the LST/LET

Either the production order finish date (customer deadline, delivery date) or the earliest end time of the production order is used to define the LST/LET.

Processing times

The total processing time is calculated for the individual operations for scheduling purposes. This total processing time is derived from the sum of the process times (durations) defined for the operations

- wait time
- setup time
- processing time/ remaining run time (see below)
- dismantling time/retooling (teardown) time
- idle time
- transport time

These process times are either set explicitly (manually or via interface) or calculated using defined formulas.

Processing time/ Remaining run time

The processing time/ remaining run time is calculated based on the remaining run time formula defined in the operation. In this context, the performance level defined for the workplace/ at the machine is taken into consideration. If the operation has not yet been planned in detail, then the performance level is used that is defined for the planned group.

During scheduling, the available capacity is considered, calculated based on the results of the shift model - year model field or otherwise the year model defined as deviating planning model at the planned [workplace](#). If the operation is not yet assigned to any individual workplace, the group shift calendar is used. Exceptions to this are transition times that - depending on the transition time - are matched to the Gregorian calendar or even to a specific shift calendar (transport time).



Please pay attention to the fact that the default values that are used for calculating the remaining run time formula are edited/maintained completely/correctly at the operation, for new operations that are included in the pool of groups and that have not yet been planned.

Production variants are not taken into account for lead time scheduling.

Affected orders

Orders that match the following criteria are scheduled with lead time:

- The order is of the order type 0, 2, 3, 5 or 6.
- The order status is prepared or started.

If these order-related criteria are met, then in continuation the following operations are considered.

Affected operations

- The operation status is prepared, running, interrupted or automatically interrupted.
- The operation is flagged based on its [processing code](#) as either relevant for scheduling or relevant for detailed planning (customized setting).

In the standard delivery, all operations receive the processing code SYSTEM. This is flagged as either relevant for scheduling or relevant for detailed planning.

General processing

The individual operations of the order are considered and started with scheduling, beginning with the first operation relevant for scheduling that is not yet finished. For operations already planned for workplaces, the planning dates are interpreted as scheduled dates (regardless of their scheduled dates), because it is assumed that the operation is processed by this date. If, on the other hand, the planning dates are set in the past, then they are ignored, i.e. the operation is rescheduled.

If the EET of an operation is later than the basic finish date of the production order, or the LST is earlier than the basic start date of the production order, then the basic finish date of the production order cannot be met. If a reduction strategy was assigned to the production order, then based on this strategy, a step-by-step attempt is now made to shorten the total duration of the order so that the basic finish date is not exceeded.

Each order is scheduled independently of the other orders. The scheduling result valid for the order (scheduled start time, scheduled end time) is defined based on the scheduling type in the order header. If no scheduling type is defined, then the scheduling type defined in the HYDRA basic settings is used.

If the order's basic start date is set in the future, then backwards scheduling sets the basic start date to "now + planning lead time". As a result, the basic start date of the order is ignored.

For operations that have the planning indicator at the processing code set to "N", processing time is treated as zero regardless of the processing times defined. In doing so, the start date of such operations is set to equal the scheduled finish date of the preceding operation and the finish date set to equal to the start date of the subsequent operation.

Planned operations are scheduled "as usual" (the planning dates are not taken into consideration). EST and LST are calculated. After scheduling is completed, the planning dates will overwrite the scheduled dates.

For running operations, the planned finish date is calculated and scheduling is continued from that time on.

Continued scheduling after OP end

If a finished operation is set in the past, the earliest start (in forward scheduling) of the successor operation is calculated based on the idle time, transport time and wait time. If the earliest start is > planning lead time, the earliest start of the successor is set to earliest start = now + planning lead time + wait time.

Results of scheduling

Results of lead time scheduling are updated dates for each operation.

The difference between LST and EST yields the buffer time for an operation. Operations with negative buffer times are deemed critical, because every delay to such an operation would result in a delay in the production order.

Calculating the order buffer or the delay when the buffer is negative:

This value is derived from the difference between the latest finish date (=requirement date) and the order's scheduled end time. Thus, the only way to make this buffer positive is by using forward scheduling. If the buffer is negative, then the order is late.

8.2 Reduction measures

It is possible to apply reduction measures as part of lead time scheduling:

If it turns out during scheduling that the lead time for a given order is longer than the allotted time available, then the system will attempt to take reduction measures to shorten the lead time accordingly.

Lead times are reduced in increments. To this end, the operation is repeated until either the deadline is met or the maximum reduction has been reached and it becomes inevitable that the deadline will be exceeded. In which steps each reduction takes place in the operation is defined by the reduction strategy that is referenced accordingly in the order header (Index tab *Dates*).

The reduction strategy can be applied to reduce the times listed below:

- Wait time, down to which, based on the steps of the reduction strategy, reductions can be made (at most: minimum value).
- Transport time, down to which, based on the steps of the reduction strategy, reductions can be made (at most: minimum value).

By scheduling using the reduction strategy described above you can reduce the lead time in order to meet the requirement date. Each result (reduction level) is recorded in the order header.

Please note:

The [reduction strategies](#) are defined during HYDRA customizing.

8.3 Executing lead time scheduling

The actions listed below cause an order to be scheduled. Keep in mind that only those orders are scheduled that meet the criteria described in [operation flow](#):

- When an order or a separate operation is created this order will run through lead time scheduling. This applies equally to orders that are transferred via interface from the ERP/ PPS system as well as to those that were created manually.
- Planning, re-planning and deallocating (removing) operations in HYDRA shop floor scheduling. Keep in mind here that scheduling does not run until after the planning has been saved.
- Modifications made to the operation manually using the operation editing functions at the client will cause the operation to be rescheduled or will initiate scheduling:
 - Modification of the workplace
 - Modification of the "external processing" flag
 - Modification of a target quantity
 - Modification of a target cycle

- Modification of process times (also by changes to parameters contained in formulas):
Lead time, wait time, setup time, additional setup time, processing time, inspection time, dismantling/teardown time, idle time, synchronization time, transport time, delivery time

In all cases, the order number of the order to which a modification was made is stored in a table (*ade_auto_verarb*).

A process runs through each of the orders listed in the table cyclically. The periodicity in which this operation is started is set in the scheduler and it should be defined based on the system.

Scheduling can also be initiated manually. The relevant functions are described [here](#).

9 Reduction Strategies

Overview

Menu	Master data → Production control → Reduction strategies
Transaction code	red
Function authorization	mdred

You can apply reduction strategies during lead time scheduling. The strategy you should use in each case is specified by the ERP system and managed in the order header.

Purpose

You can apply reduction strategies during lead time scheduling. The strategy you should use in each case is specified by the ERP system and managed in the order header.



This function is only useful, and thus relevant if you schedule orders in HYDRA.

Field descriptions

Reduction strategy

Reduction strategy: strategy key (referenced in the order header).

Reduction level

Reduction level. Levels are processed in ascending order.

Reduction of waiting time

The amount of time, expressed in percent, by which waiting time is reduced at this level.

Reduction of transport time

The amount of time, expressed in percent, by which transport time is reduced at this level.

Comment

Comment or description

10 Transport Matrix

Summary

Menu	Master Data → Production Control → Transport Matrix
Transaction code	ttx
Function authorization	mdttx

A transport matrix can be defined to determine the transport time between two operations.

Usage

When a new operation is created or when an explicit workplace or group change is made manually using the operation update function, the transport time is determined using this matrix and the results are transferred into the operation. Any change to the transport matrix or any rescheduling in the graphic planning board later will have no effect on already existing operations.



This function is only meaningful and therefore relevant if order scheduling is run in HYDRA.

In order to reduce the amount of data when determining the transport time from one workplace to another, the workplaces should be assigned to so-called location groups first. To do this, create a group in the configuration application [Groups](#) with the identifier "Location group" and use the configuration application [Group assignment](#) to assign the location group to a workplace.

Field descriptions

From location group

Location group of the original workplace

To location group

Location group of the target workplace

Transport time - normal

Normal transport time in hours.

Transport time - minimum

Minimum transport time in hours, down to which, based on the steps of the reduction strategy, reductions can be made.

Calendar

With regard to the calendar that scheduling is based on, the following options are available:

- G = Gregorian calendar
Transport times are scheduled using the Gregorian calendar.
- S = Shift calendar
When this option is set, the shift calendar or the original workplace is used for scheduling.
- T = Shift model from transport matrix
The shift model entered in the field Shift model is used for scheduling.

Shift model

Shift model that should be used when the option "T" is set in the calendar field.

Please note: The number of different shift models that are defined here should be kept to a minimum, because an increased number of different shift models may adversely impact performance in HYDRA shop floor scheduling.

Comment

Comment about this entry